C4 Energetics – Knowledge Organiser: Self-Test

AQA A-level Chemistry • Set 1: Key Definitions **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Enthalpy change (ΔH)				Heat energy change at constant pressure
2	Sign of ΔH for exothermic reactions				Negative
3	Sign of ΔH for endothermic reactions				Positive
4	Standard conditions				100 kPa and 298 K
5	Standard enthalpy of combustion ($\Delta_c H^{\ominus}$)				Enthalpy change when 1 mol burns completely in oxygen, under standard conditions, all in standard states
6	Standard enthalpy of formation ($\Delta_f H^{\ominus}$)				Enthalpy change when 1 mol of a compound forms from its elements under standard conditions, with **elements and product in standard states**
7	$\Delta_f H^{\ominus}$ of an element in its standard state				0
8	Mean bond enthalpy				Energy to break 1 mol of a bond in gaseous molecules, averaged over many compounds
9	Bond breaking: endothermic or exothermic				Endothermic
10	Bond making: endothermic or exothermic				Exothermic

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 2: Equations **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Calorimetry equation				$q = mc\Delta T$
2	Units of q in $q = mc\Delta T$				J
3	Specific heat capacity of water				4.18 J g^{-1} K^{-1}
4	Equation for ΔH from q				$\Delta H = -q \div n$
5	Units of ΔH				kJ mol^{-1}
6	ΔH from bond enthalpies				Σ bonds broken − Σ bonds made
7	ΔH from enthalpies of formation				Σ $\Delta_f H$ products − Σ $\Delta_f H$ reactants
8	ΔH from enthalpies of combustion				Σ $\Delta_c H$ reactants − Σ $\Delta_c H$ products
9	Hess's law				Enthalpy change is independent of the route taken
10	Why mean bond enthalpy values are less accurate				Averaged over different compounds

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.

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AQA A-level Chemistry • Set 3: Calorimetry and Standard States **Papers 1 & 2**

#	Question	Attempt 1	Attempt 2	Attempt 3	Answer
1	Main source of error in calorimetry				Heat loss to surroundings
2	Other error in combustion experiments				Incomplete combustion
3	What m is in $q = mc\Delta T$				Mass of water or solution heated
4	Mass of 1 cm^3 of aqueous solution				1 g
5	Convert J to kJ				÷ 1000
6	Symbol for standard conditions				$^{\ominus}$
7	Standard state of carbon				C(s, graphite)
8	Standard state of oxygen				$O_2(g)$
9	Standard state of bromine				$Br_2(l)$
10	Standard state of water				$H_2O(l)$

Fold the Answer column behind the page. Answer each question from memory, then unfold and check. Repeat on a later day in the next Attempt column.